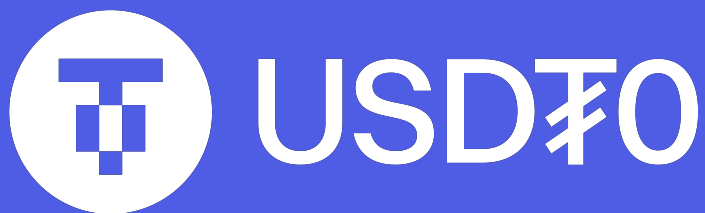


Everdawn USAT XAUTO Celo Deployment Review



April 29, 2026

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Summary

Type	Stablecoins	Total Issues	2 (0 resolved)
Timeline	From 2026-04-09 To 2026-04-09	Critical Severity Issues	0 (0 resolved)
Languages	Solidity	High Severity Issues	1 (0 resolved)
		Medium Severity Issues	0 (0 resolved)
		Low Severity Issues	0 (0 resolved)
		Notes & Additional Information	1 (0 resolved)

Scope

This report provides a detailed assessment of the on-chain deployment of the USAT token and its LayerZero OFT contracts, as well as the proposed upgrade of XAUT0 on the Celo network.

The deployment was introduced via [pull request #100](#) of the [Everdawn-Labs/usdt0-tether-contracts-hardhat](#) repository. Additionally, the deployment and wiring scripts introduced via [pull request #205](#) were reviewed.

System Overview

The deployment assessment process consists of a comprehensive series of steps designed to confirm the on-chain correctness of all contracts and their initializations.

The review focused on the deployment of USAT on Celo, the proposed upgrade of XAUT0 on Celo to support Celo Fee Abstraction via `CeloOFTExtension` and `FeeCurrencyWrapper` contracts, and the USAT OFT deployment and LayerZero wiring configuration across Celo and Ethereum. The following items were verified as part of the assessment:

1. Token, ProxyAdmin, and OFT deployment parameters and contract ownership.
2. Proxies point to their intended implementation addresses.
3. ProxyAdmin contracts are properly configured and associated with the respective proxies.
4. The OFT contract is linked to the correct token address and the appropriate LayerZero endpoint configuration for the given chain.
5. Where applicable, scheduled transactions in the Safe related to the setup and configuration of these contracts are reviewed upon request.
6. Pending upgrade implementation (`CeloOFTExtension`) is deployed and ready for XAUT0.
7. Registration of FeeCurrencyWrappers in Celo's fee currency registry.
8. Implementation bytecode matches compiled source from the repository.
9. Peer associations between OFT and OAdapter contracts.
10. LayerZero wiring configuration (DVNs, block confirmations, enforced options) compared against existing XAUT0 production config.

A security review was also performed of the following contracts in [PR#100](#):

- `TetherTokenCeloExtension` — Extends `TetherTokenV2` with `debitGasFees` / `creditGasFees` functions and a `feeCurrencyWrapper` access control mechanism. Existing contract with superficial changes.
- `CeloOFTExtension` — New contract inheriting from both `TetherTokenOFTExtension` and `TetherTokenCeloExtension`, resolving diamond inheritance conflicts for `name`, `symbol`, and `_EIP712NameHash`. Used as the implementation for both XAUT0 (pending upgrade) and USAT (fresh deployment).

The `USATDeployments.ts` and `layerzero-usat.config.ts` scripts were also reviewed, in addition to updating the evm version in `hardhat.config.ts` to "shanghai".

Note: `FeeCurrencyWrapper` existed prior to this PR and is considered out of scope for the contract audit. Its deployment configuration is verified in the deployment assessment below.

Additionally, a review was performed of the LayerZero wiring and deployment scripts from the [Everdawn-Labs/usdt0-oft-contracts](#) repository, covering `USATDeployments.ts`, `layerzero-usat.config.ts`, and `hardhat.config.ts`. The USAT OFT (`OUpgradeable`) on Celo and OAdapter (`OAdapterUpgradeable`) on Ethereum were verified on-chain against the deployment script parameters, including token linkage, endpoint configuration, proxy admin ownership, and initialization. The LayerZero wiring configuration was compared against the existing XAUT0 production config on the same pathways, covering DVN selection, block confirmations, and enforced options.

All USAT contracts — token, FeeCurrencyWrapper, OFT, and OAdapter — along with their proxy admins, have had ownership transferred to Safe multisig contracts (`0x8caD...E1BC` on Celo, `0x62b3...5Cb` on Ethereum), each with 5 owners. XAUT0 contracts remain owned by the existing Safe multisig at `0x4DFF...0bf8`.

More information can be found in the sections below.

On-Chain Deployments

The on-chain deployment verification is organized into two categories:

Token and FeeCurrencyWrapper contracts (Tables 1–4): Deployed via [PR#100](#) in the `usdt0-tether-contracts-hardhat` repository. These cover the XAUT0 upgrade and USAT token deployment on Celo, along with their respective FeeCurrencyWrapper contracts.

OFT and OAdapter contracts (Tables 5–6): Deployed via [PR#205](#) in the `usdt0-oft-contracts` repository. These cover the USAT OFT on Celo (`0Upgradeable`) and the USAT OAdapter on Ethereum (`0AdapterUpgradeable`), which handle LayerZero cross-chain messaging.

1. XAUT0 Token (Upgrade)

Check	Expected	Status
Proxy address	<code>0xaf37...47ff</code>	✓
Current implementation	<code>Celo0FTExtension</code>	✗ — Still <code>TetherToken0FTExtension</code> (<code>0xe7cd...</code>). Upgrade queued, not executed.
Pending implementation bytecode	Matches compiled <code>Celo0FTExtension</code>	✓ — <code>0x2070...7e4C</code>
Proxy admin	<code>0x01bf...1071</code>	✓
Proxy admin owner	<code>0x4DFF...0bf8</code>	✓
Token owner	<code>0x4DFF...0bf8</code>	✓
Name	<code>XAUt0</code>	✓
Symbol	<code>XAUt0</code>	✓

Check	Expected	Status
Decimals	6	✓
OFT contract	0x21c...A3bf	✓
LZ endpoint (on OFT)	0x1a44...728c	✓

2. XAUT0 FeeCurrencyWrapper

Check	Expected	Status
Proxy address	0x857B...4C16	✓
Implementation bytecode	Matches compiled FeeCurrencyWrapper	✓ — 0x06d8...8e9a
Proxy admin	0x7B41...285C	✓
Proxy admin owner	0x4DFF...0bf8	✓
Wrapped token	0xaf37...47ff (XAUT0 proxy)	✓
Registered in Celo fee registry	Registered	⚠ Pending — requires Celo governance proposal to register in FeeCurrencyDirectory

3. USAT Token (Fresh Deployment)

Check	Expected	Status
Proxy address	0xD2ab...F771	✓
Implementation bytecode	Matches compiled CeloOFTExtension	✓ — 0x4991...b71E
Proxy admin	0x407c...7422	✓
Proxy admin owner	Safe multisig	✓ — 0x8caD...E1BC

Check	Expected	Status
Token owner	Safe multisig	✓ — 0x8caD...E1BC
OFT contract	0xC8F4...d0E	✓
Name	Tether America USD	✓
Symbol	USAT	✓
Decimals	6	✓

4. USAT FeeCurrencyWrapper

Check	Expected	Status
Proxy address	0x0357E...80C4	✓
Implementation bytecode	Matches compiled FeeCurrencyWrapper	✓ — 0xACc6...f5F4
Proxy admin	0x407c...7422	✓
Proxy admin owner	Safe multisig	✓ — 0x8caD...E1BC
Wrapped token	0xD2ab...F771 (USAT proxy)	✓
Registered in Celo fee registry	Registered	⚠ Pending — requires Celo governance proposal to register in FeeCurrencyDirectory

5. USAT OFT on Celo

Check	Expected	Status
Proxy address	0xC8F4...d0E	✓
Implementation	0x338d...fce	✓
Proxy admin	0xbfd9...b80	✓

Check	Expected	Status
Proxy admin owner	Safe multisig	✓ — 0x8caD...E1BC (5 owners)
Owner	Safe multisig	✓ — 0x8caD...E1BC
Token	0xD2ab...F771 (USAT proxy)	✓
LZ endpoint	0x1a44...728c	✓
Peer to Ethereum (30101)	0x1817...43DF (OAdapter)	✓

6. USAT OAdapter on Ethereum

Check	Expected	Status
Proxy address	0x1817...43DF	✓
Proxy admin	0xd122...b35f	✓
Proxy admin owner	Safe multisig	✓ — 0x62b3...5Cb (5 owners)
Owner	Safe multisig	✓ — 0x62b3...5Cb
Token	0x0704...8b68 (USAT on Ethereum)	✓
LZ endpoint	0x1a44...728c	✓
Peer to Celo (30125)	0xC8F4...d0E (OFT)	✓

Deployment parameters and contract ownership were established correctly in this PR, with ownership transferred to Safe multisig contracts on both chains. Peer associations are correctly configured in both directions between the Celo OFT and Ethereum OAdapter.

Note that the `upgradeTo` call on XAUT0 was not yet executed, in order to upgrade the implementation from `TetherTokenOFTExtension` to the new `CeloOFTExtension`. It has been queued as a [Safe transaction](#) (nonce 54), and the calldata and target is correct for the

upgrade to successfully execute. Additionally, the `setFeeCurrencyWrapper` function has been queued to be called on the XAUT0 contract after it's upgraded, in a separate [Safe transaction](#) (nonce 55) that is also correctly structured. Finally, both newly deployed `FeeCurrencyWrapper` contracts should be registered in the [FeeCurrencyDirectory](#), done by Celo governance.

A [forum post](#) proposing to enable both USA $\text{\textcircled{C}}$ and XAU $\text{\textcircled{C}}$ 0 as gas currencies on Celo has been published, but no formal on-chain governance proposal was found at the time of this review.

High Severity

H-01 Endpoint ID differs from intended implementations

In `layerzero-usat.config.ts`, the `eid` field of `celoContract` is set to `EndpointId.CONFLUX_V2_MAINNET`. However, Celo's endpoint id is 30125 and Conflux's endpoint id 30212. Given the difference in endpoint ids, the Celo oft contract would be peered to Conflux endpoint ID rather than Celo. Cross chain messages from Ethereum would route to Conflux and messages from Celo would never be received since the peer association points to the wrong chain. USAT cross chain transfers would be completely non functional. Since this is a pre-deployment config, and the wiring hasn't been applied yet, it can be caught during wiring if tooling checks endpoint support, but if executed unnoticed it breaks all cross chain functionality for USAT.

Consider setting the `eid` field to `EndpointId.CELO_V2_MAINNET`.

Notes & Additional Information

N-01 Inconsistent Config Between XAUT0 and USAT

The block confirmations are set much higher for USAT than XAUT0 on both mainnet and Celo. Gas requirements also differ between what was set for USAT and XAUT0. - XAUT0 block confirmations set to 1500 and 15 on celo and eth, respectively. `msgType2 LZ_RECEIVE` and `COMPOSE` gas both set to 100,000 for XAUT0. - USAT block confirmations set to 2500 and 96 on celo and eth, respectively. USAT config has 200,000 for everything, including `msgType2` which differs from what XAUT0 uses.

Higher confirmations just mean longer finality wait times for users. If deployment is done using these more conservative values, USAT transfers will be noticeably slower without a clear

reason. Higher gas also means users pay slightly more per message, just another inconsistency alongside block confirmations.

Consider maintaining consistent configuration between tokens deployed on the respective chains.

Conclusion

The review of [PR#100](#) and [PR#205](#) covered the deployment of USAT on Celo, the proposed upgrade of XAUT0 on Celo to [CeloOFTExtension](#), the deployment of [FeeCurrencyWrapper](#) contracts for both tokens to support Celo Fee Abstraction, as well as the USAT OFT and OAdapter deployments on Celo and Ethereum respectively.

Two issues were identified with the intended configurations, relating to the LayerZero wiring configuration of the USAT OFT. We thank the team for clear communication in providing all needed context.

Appendix

Issue Classification

OpenZeppelin classifies smart contract vulnerabilities on a 5-level scale:

- Critical
- High
- Medium
- Low
- Note/Information

Critical Severity

This classification is applied when the issue's impact is catastrophic, threatening extensive damage to the client's reputation and/or causing severe financial loss to the client or users. The likelihood of exploitation can be high, warranting a swift response. Critical issues typically involve significant risks such as the permanent loss or locking of a large volume of users' sensitive assets or the failure of core system functionalities without viable mitigations. These issues demand immediate attention due to their potential to compromise system integrity or user trust significantly.

High Severity

These issues are characterized by the potential to substantially impact the client's reputation and/or result in considerable financial losses. The likelihood of exploitation is significant, warranting a swift response. Such issues might include temporary loss or locking of a significant number of users' sensitive assets or disruptions to critical system functionalities, albeit with potential, yet limited, mitigations available. The emphasis is on the significant but not always catastrophic effects on system operation or asset security, necessitating prompt and effective remediation.

Medium Severity

Issues classified as being of medium severity can lead to a noticeable negative impact on the client's reputation and/or moderate financial losses. Such issues, if left unattended, have a moderate likelihood of being exploited or may cause unwanted side effects in the system.

These issues are typically confined to a smaller subset of users' sensitive assets or might involve deviations from the specified system design that, while not directly financial in nature, compromise system integrity or user experience. The focus here is on issues that pose a real but contained risk, warranting timely attention to prevent escalation.

Low Severity

Low-severity issues are those that have a low impact on the client's operations and/or reputation. These issues may represent minor risks or inefficiencies to the client's specific business model. They are identified as areas for improvement that, while not urgent, could enhance the security and quality of the codebase if addressed.

Notes & Additional Information Severity

This category is reserved for issues that, despite having a minimal impact, are still important to resolve. Addressing these issues contributes to the overall security posture and code quality improvement but does not require immediate action. It reflects a commitment to maintaining high standards and continuous improvement, even in areas that do not pose immediate risks.